Aeroacoustic Flight Test and Data Analysis for the Validation of Fenestron Noise Computations

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Fabrice Falissard
The French Aerospace Lab ONERA

Fabrice Desmerger
O.N.E.R.A

Laurent Binet
The French Aerospace Lab ONERA

Aeroacoustic Flight Test and Data Analysis for the Validation of Fenestron Noise Computations

F. Falissard*, F. Desmerger*, P. Gardarein**, L. Binet***, J.-C. Camus⋆

* CFD and Aeroacoustics Department
ONERA, Châtillon, France

** Applied Aerodynamics Department
ONERA, Meudon, France

*** Control and Flight Dynamics Department
ONERA, Salon de Provence, France

⋆ CEV Istres
DGA Flight Test, Istres, France

Corresponding author: Fabrice.Falissard@onera.fr

The Fenestron Noise Study, done in partnership by ONERA, Eurocopter and DGA Flight Test, focuses on the noise radiated by a Fenestron shrouded tail rotor depending on the flight conditions. This study combines test measurements, numerical simulations and has two main objectives: first to enhance the comprehension of the noise generation mechanisms involved in a shrouded tail rotor, secondly to validate the numerical tools developed at ONERA for the calculation and prediction of helicopter aerodynamics and aeroacoustics. This paper presents the experimental setup, the flight tests and the data processing method dedicated to the validation of the simulation tools. The second part of the paper describes the simulation tools used for aerodynamic and aeroacoustic predictions of a Fenestron tail rotor. Finally, preliminary comparisons between experiments and calculations are shown for a level flight at speed 150 kts.

Notation

BPF	Blade Passing Frequency
CAA	Computational AeroAcoustics
CEV	Centre d'Essais en Vol
CFD	Computational Fluid Dynamics
DGA	Délégation Générale de l'Armement
DGAC	Direction Générale de l'Aviation Civile
(D)GPS	(Differential) Global Positioning System
elsA	ensemble logiciel pour la simulation en aérodynamique
FW-H	Ffowcs Williams–Hawkings
HOST	Helicopter Overall Simulation Tool
ONERA	Office National d'Etudes et de Recherches Aérospatiales
SPL	Sound Pressure Level
TAS	True Air Speed
TSA	Time Synchronous Averaging
URANS	Unsteady Reynolds Averaged Navier-Stokes

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Introduction

Quiet helicopter studies for environmental concerns are intended to reduce rotorcraft external noise. This external noise results from the contribution of three main sources which are the main rotor, the tail rotor and the engines. The ranking of these different sources on the radiated noise strongly depends on the flight condition and on the observer location so that they all have to be reduced. Although major improvements have been achieved in noise reduction during the last decades there is still a need to further improve the noise reduction for a better acceptance of helicopter operations in populated areas.

In this context a specific study dedicated to Fenestron noise has been launched in 2004 by ONERA and Eurocopter with support of the French Ministry of Civil Aviation (DGAC). This study has two main objectives, first to acquire an experimental comprehensive database dedicated to Fenestron noise analysis, secondly to validate the numerical tools developed at ONERA for Fenestron aerodynamic and aeroacoustic predictions. The experimental program of the study is divided in two phases. The first phase consists in flight tests with embedded instrumentation for in-flight aerodynamic and acoustic measurements only. The objective of this first flight test campaign was to test as many flight conditions as possible to gather an extensive database for aerodynamic and acoustic analysis. The second phase consists in flight tests with ground noise measurements for flight conditions selected for their interest in terms of acoustic certification or specific operating points of the Fenestron identified during the first flight test campaign.

Several results of the Fenestron Noise Study have been presented in previous communications; the description and development of the specific instrumentation designed for aerodynamic and acoustic in-flight measurements are detailed in Refs. 1,2; the results of the first flight test campaign exhibiting strong correlations between the flow characteristics in the Fenestron duct and acoustic levels in the near field are discussed in Ref. 3; the results and the analysis of the numerical simulations performed for several flight conditions are presented in Refs. 4, 5 and comparisons between numerical simulations and in-flight measurements are given in Ref. 6.

This paper addresses the aeroacoustic flight test and data analysis for the validation of Fenestron noise computation with a specific emphasis on the second flight test campaign with ground noise measurements. It is organized as follows: first the experimental setup is presented; then the test matrix of both test campaigns are summarized; the third section deals with the post-processing applied to the measured data for the validation of numerical simulations; the following section presents the numerical tools and the strategy used at ONERA for Fenestron noise prediction; finally aerodynamic and aeroacoustic computations are compared to the experiments for a high speed level flight condition.

Experimental setup

The experimental setup used for the Fenestron Noise Study consists in two parts. The first part is the Dauphin tail number 6075 from the DGA Flight Test Center (CEV) that has been highly instrumented for these experiments (Refs. 1, 2). The second part is made of ONERA's acoustic ground instrumentation used for flyover noise measurements. Two test campaigns have been carried out, the first one dedicated to in-flight measurement only with a highly instrumented helicopter and many flight conditions, the second one dedicated to ground noise measurements involving the same helicopter but with a lighter in-flight instrumentation.

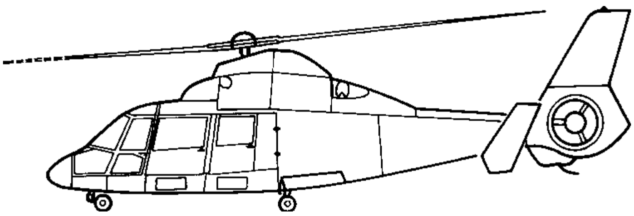

Fig. 1. View of the SA 365 N Dauphin 2 helicopter

Test aircraft

The Dauphin tail number 6075 is an Eurocopter SA 365 N Dauphin 2 helicopter. This aircraft, displayed in Fig. 1, is equipped with a 4 bladed main rotor (rotating clockwise when seen from the top) and with a 13 bladed Fenestron shrouded tail rotor. This Fenestron is of first generation (Ref. 7) and very close in its design to the Fenestron of the SA 341 and SA 342 Gazelle helicopters (Ref. 8). As can be seen in Fig. 2, the rotor has 13 equally spaced blades and is rotating clockwise when seen from port or main rotor advancing blade side. The rotor is mounted on a hub supported by 3 equally spaced arms, two of these arms possess a biconvex section while the last arm possesses a larger D-section in order to shelter the transmission shaft. The shroud diameter is 0.9 m and the nominal rotation speed is 4700 rpm providing a blade passing frequency (BPF) close to 1 kHz.

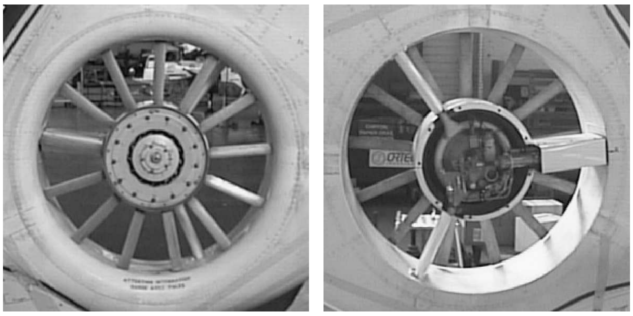

Fig. 2. Port and starboard views of the SA 365 N Dauphin 2 Fenestron tail rotor (without hub casing)

Obviously this basic geometry is quite different from modern Fenestrons with uneven blade spacing for acoustic concerns (Refs. 9–11) but this simpler design is thought to be an advantage for numerical simulation and physical analysis. The characteristic of the main and tail rotor are summarized in Tab. 1

Table 1. SA 365 N Dauphin 2 helicopter main and tail rotor characteristics

	Main rotor	Tail rotor
Number of blades	4	13
Diameter (m)	11.93	0.9
Rotation speed (rpm)	350	4706
BPF (Hz)	23	1020

Aircraft instrumentation

The development, integration and testing of the specific instrumentation used for the in-flight measurements has been fully described in previous papers (Refs. 1, 2). It consists in two distinct parts, one fixed on the aircraft fuselage and one rotating with the tail rotor.

The fixed instrumentation, displayed in Fig. 3, is composed of steady pressure sensors on the rear part of the fuselage and tail, of steady and unsteady pressure sensors on the Fenestron casing, of Pitot probe rakes in the duct and of 1/4" microphones fixed on the horizontal stabilizers. The acoustic signals provided by the 4 microphones were sampled at 80 kHz.

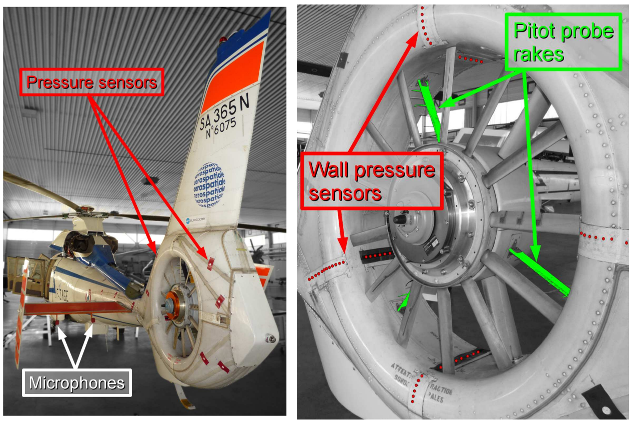

Fig. 3. Specific instrumentation used for in-flight aerodynamic and acoustic measurements

The rotating part of the instrumentation, presented in Fig. 4, consists in a trigger system, in pair of upper/lower thin layer unsteady Kulite pressure transducers mounted on 4 blades of the rotor at 0.7 radius and in strain gauges fixed on a reference blade. The Kulite and gauge signals were wirelessly transmitted to the recording system by means of a dedicated telemetry system. The unsteady blade pressure signals were sampled at 20 kHz which is sufficient considering a rotation frequency of 80 Hz.

In order to post-process and analyze the data, flight parameters such as true air speed, vertical velocity, yaw, pitch and roll angles were also recorded during the flight test.

The specific instrumentation mounted on the aircraft added some meteorological constraints for the flight test such as low-humidity and no-rain to preserve the microphones or outside temperature lower than 30 °C, so that the glue used to fix the silicon strip holding back the steady pressure sensor would stick strongly on the fuselage.

Between the two test campaigns, the helicopter had to be used by the CEV for other purposes, providing time to perform some repairs and modifications of the instrumentation. Nevertheless, many problems concerning the telemetry occurred at the beginning of the second campaign so that the rotating part of the instrumentation had to be removed for the ground noise measurements and the instrumented rotor was replaced by a classical one.

Finally, in order to better fit the numerical simulation with the experiments, the tail rotor torque and blades pitch angle were measured during the ground noise measurement.

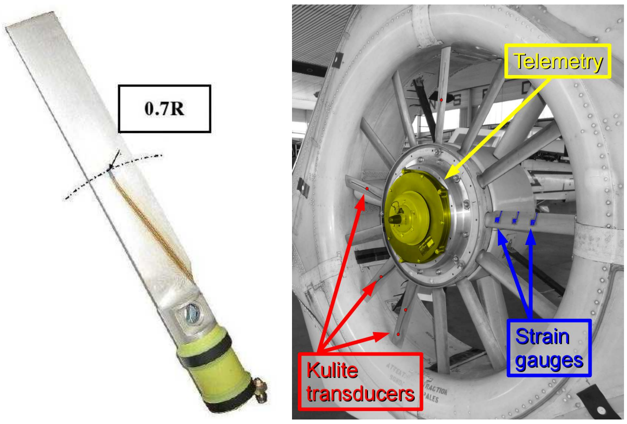

Fig. 4. Specific rotating instrumentation used for in-flight blade pressure measurements

Test site for ground measurements

The French Air Force Base of Istres, where the CEV is located, was chosen as test site. Although Istres is a large multi-role tasked Air Force base with several operational units and some test facilities, the heavy traffic was not a major constraint during noise measurements because regulation of the air traffic was made to limit the background noise as far as possible.

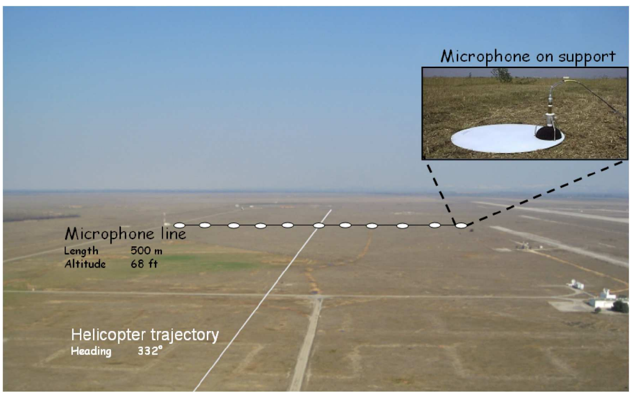

Fig. 5. Test site for ground noise measurements

Microphone array for ground measurements

The microphone array used for ground noise measurements is made up of 11 ground-mounted microphones equally spaced on a line of 500 m length. The microphone line has been oriented in such a way that the nominal flight path be parallel to the usual wind direction on the test site. In order to know the relative position and orientation of the helicopter with respect to the microphones, they were precisely geo-positioned after installation.

The 1/2” B&K microphones where mounted head down on dedicated tripod struts, see Fig. 6, so that the microphone membrane is placed 7 mm above a metallic plate of 0.5 m diameter. In this way, reflected and incident noise are almost identical on a wide range of frequencies and incidence angles, and the measured pressure can be assumed to be twice the incident pressure. Standard 4 inch wind screens were used. The acoustic signals and time signal provided by a GPS antenna were recorded using ONERA’s new digital acoustic measurement system based on B&K LAN-XI data acquisition hardware. The microphones were calibrated each measurement day morning using a 94 dB–1000 Hz pistonphone. During the tests, the acoustic signals were sampled at 131 kHz.

Since the test site was located on a military base of the French Air Force, all the ground test measurement systems had to be removed every evening and repositioned every measurement day. The time necessary for a complete installation and testing of the system was approximately one and half hour.

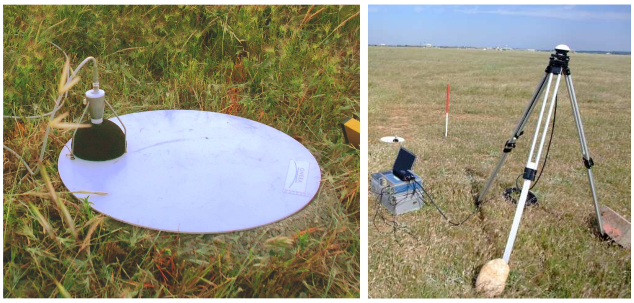

Fig. 6. Microphone setup (left) and GPS receiver used for time recording (right)

Since the helicopter trajectory had to be perpendicular and centered on the microphone line (see Fig. 5), a specific device was used by the pilots in order to help them to always follow the same ground trajectory. A Flight Director, based on the Dauphin onboard instrumentation was designed in order to display the approach profile to the pilot. The device was based on the computation of heading (track) and slope angle deviations from reference points located on the glide path. These reference points were selected in order to provide an accurate trajectory.

For descent or take-off scenarios presented in Fig. 7 and 9, two reference points were defined for each of the 2 segments composing the trajectory so that the pilot had to maintain a level flight up to the first reference point and then had to intercept and maintain the glide path up to the second reference point behind the microphones line.

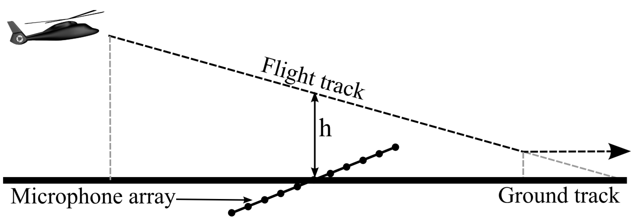

Fig. 7. Outline of approach profile

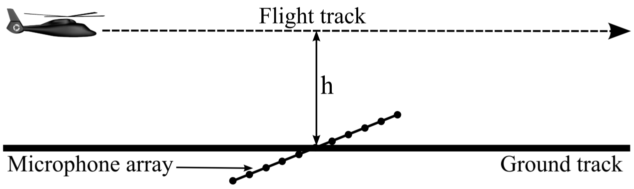

Fig. 8. Outline of level flight profile

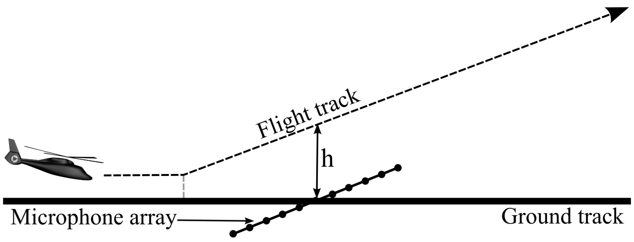

Fig. 9. Outline of take-off profile

Tracking and guidance for ground measurements

In addition to the basic on-board instrumentation, a GPS was used to precisely measure the flight path of the helicopter. As a DGPS system was not available in flight, signals were post-treated to be synchronized with the ground GPS platform and acoustic signals.

The glide path indication was given to the pilot through a specific instrument onboard the Dauphin as presented in Fig. 10. The crossed needles on this display were used to give the helicopter slope angle and heading. The objective for the pilot was to keep the needles intersection at the screen centre in order to remain on the targeted glide slope and flight direction.

Fig. 10. Crossed needles used for flight path guidance

Weather data

The weather data were provided by the airport weather station. The recorded data consist in the temperature, the relative humidity, the wind direction and the wind speed (mean and max values during the last minute) measured at 82 ft.

Test matrix

Different test programs have been defined for each test campaign with flight conditions chosen by ONERA and Eurocopter in order to build up an extensive data base that will be first used to understand the aerodynamic and acoustic phenomena occurring in the Fenestron depending on the flight parameters and secondly to validate the computational methods and sofwares developed at ONERA for helicopter aerodynamic and aeroacoustic predictions.

Test matrix for in-flight measurements

The first test campaign has been carried out in 7 flights by the CEV crew from November 2008 to January 2009 and consists in many stabilized flight conditions including level flights, take-off and approaches. The different flight conditions are summarized in Tab. 2 to 4. As can be seen, many flight conditions have been repeated to assess the variability of the data depending on the test day or on the flight condition deviation.

Test matrix for ground measurements

Previously planned to begin in September 2009, the second test campaign finally started at the end of March 2010 and the test flights were performed at the end of May. This campaign was successfully achieved in 4 flights and 5.5 flight hours. All the flight conditions measurement during these tests are summarized in Tab. 5 to 7. As can be seen, there are much less measured points than in the first campaign. The test points have been selected based on the feedback and analysis of the first flight tests for their interest in terms of acoustic nuisance and flow characteristics as well as to account for noise certification configurations.

Table 2. Level flight test matrix for in-flight measurements

V_{TAS} (kts) / V_Z (ft/mn)	50	60	75	90	105	120	135	150
0	2	3	4	1	2	3	3	3

Table 3. Approach test matrix for in-flight measurements

V_{TAS} (kts) / V_Z (ft/mn)	50	60	75	90	105	120
-250	1	1	2	2	1	1
-500	3	2	2	1	1	3
-800	6	1	6	7	2	1
-1000	1	2	2	4	1	1
-1250	4	3	4	2	1	1
-1500	1	3	5	1	1	2

Table 4. Takeoff test matrix for in-flight measurements

V_{TAS} (kts) / V_Z (ft/mn)	55	75	90
250	2	1	1
500	2	3	3
750	3	5	4
1000	1	1	3
1500	2	2	2
V_{Zmax}	3	4	3

Table 5. Level flight test matrix for ground measurements

V_{TAS} (kts) / V_Z (ft/mn)	50	60	75	90	105	120	135	150
0	3	3	3	3	3	4	3	4

Table 6. Approach test matrix for ground measurements

V_{TAS} (kts) / V_Z (ft/mn)	50	60	75	90	105
-500	ø	1	1	1	2
-800	ø	1	5	2	1
-1000	3	2	4	2	1
-1250	ø	2	3	2	ø
-1500	ø	ø	5	ø	ø

Table 7. Takeoff test matrix for ground measurements

V_{TAS} (kts) / V_Z (ft/mn)	55	75	90
V_{Zmax}	4	3	3

Data processing

Different post-processing have been applied to the experimental data depending on whether the purpose was to acquire physical insight and analysis on Fenestron noise generation mechanism or computational aerodynamic and aeroacoustic code validation. Since this paper mainly focuses on data analysis for Fenestron noise computation validation, only post-processing dedicated to code validation are presented in this section.

In-flight data post-processing

Before applying any processing to the data, a time interval of the recording was selected for each test point in order to minimize the variation of the flight parameters such as flight velocity, climb and descent rate and side-slip angle (see Fig. 11). The same processing was applied to the on-board microphone and blade pressure signals. It is based on a Time Synchronous Averaging (TSA) of the unsteady signals using the experimental trigger synchronized with the rotation of the tail rotor. This process eliminates broadband noise as well as any tonal noise that is not strictly synchronous with the Fenestron rotation. This processing, classically used for rotorcraft analysis, returns spectra with very low background noise and well-defined tones as can be seen in the examples of synchronous averages of the microphone signals and unsteady blade pressures for a take-off at 75 kts and 500 ft/mn displayed Fig. 12 and 13. All test points of the first flight test campaign have been analyzed using this method, exhibiting strong correlations between unsteady blade pressure fluctuations and radiated noise at microphone locations depending on flight conditions (Ref. 3).

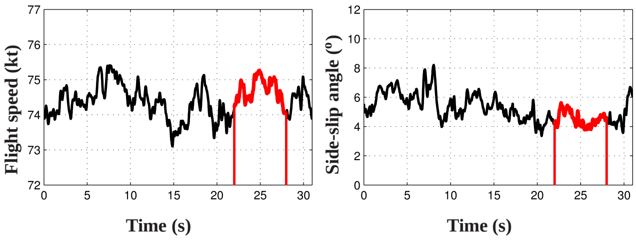

Fig. 11. Example of flight parameter variations and time interval selection for in-flight data analysis

Ground measurement data processing

The microphone signals are processed in order to compute ground noise footprints that will be used for comparison with numerical computations. Since the CFD and CAA computations are carried out in a wind tunnel reference frame, the Doppler effect occurring during the test measurements has to be compensated.

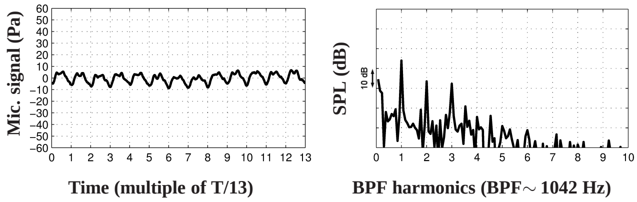

Fig. 12. Time synchronous average of microphone signal (left) and corresponding spectrum (right)

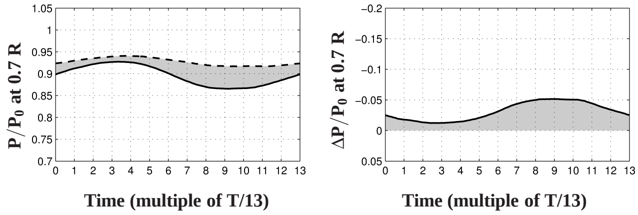

Fig. 13. Time synchronous average of the blade upper and lower side unsteady pressure signals (left) and of the upper/lower pressure difference (right)

The de-dopplerization of the recorded acoustic signals is performed the same way as for noise source localization (Ref. 12). The propagation medium is assumed to be at rest with sound velocity c. The sound pressure emitted at time t_E by a point source S on the helicopter will be received on the m^{th} microphone of the array at the reception time t_R satisfying the relation:

$$t_R = t_E + \tau_m^S(t_E)$$

where $\tau_m^S(t_E) = R_m^S(t_E)/c$ is the propagation delay between the moving source S and the fixed m^{th} microphone with $R_m^S(t_E)$ the distance between the source S and the m^{th} microphone at the emission time t_E. The de-dopplerized temporal signal $p_m^{\text{dDop}}(t_R)$, output of the m^{th} microphone at reception time t_R is then:

$$p_m^{\text{dDop}}(t_R) = p_m^{\text{dDop}}\left(t_E + \tau_m^S(t_E)\right)$$

In practice, this time de-dopplerization method requires knowing the helicopter flight path $R_m^S(t_E)$ at emission time to compute the time delay $\tau_m^S(t_E)$ and a common time reference to synchronize the in-flight data recorded at emission time and with the ground data recorded at reception time. Once the acoustic signals have been de-dopplerized, the microphone recordings are divided in data blocks related to the aircraft positions as shown in Fig. 14 providing noise levels and spectra as a function of the microphone locations. The ground noise footprint is then obtained by freezing the aircraft position and projecting the measured noise directivities onto the ground plane as displayed in Fig. 15. To obtain footprint directly comparable with CFD and CAA results, the measured ground footprint is transposed to a wind tunnel reference frame by modifying the noise directivities accounting for the flight speed as shown in Fig. 16.

The footprint related to Fenestron noise is performed by extracting and summing the tones corresponding to the first harmonics of the tail rotor. This operation is easily feasible thanks to the de-dopplerization and knowledge of the rotation speed measured during the test, combined with the fact that this tail rotor has a Blade Passing Frequency much higher than the main rotor (1000 Hz compared to 24 Hz) so that the tones related to the tail rotor are well separated from the tones related the main rotor harmonics.

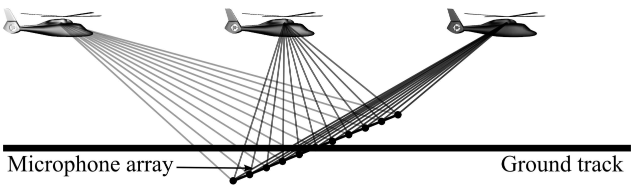

Fig. 14. Acoustic measurements during flyover

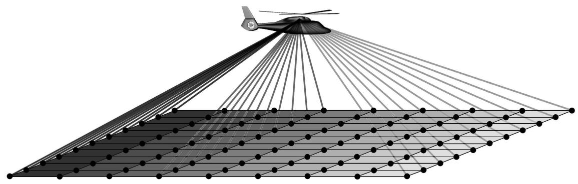

Fig. 15. Single source transformation

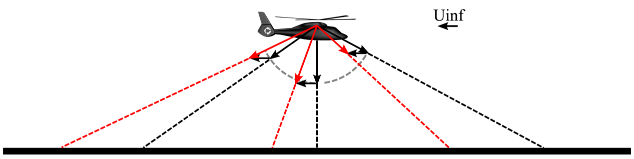

Fig. 16. Footprint transposition to the CFD/CAA reference frame

Computational method

The computational method used to calculate the noise radiated by a Fenestron tail rotor is based on the combination of CFD to compute the generation and propagation of the acoustic sources in the shroud and of integral acoustic methods to propagate these sources in the far-field (Ref. 6).

Aerodynamic computations

The aerodynamic flow over the helicopter is computed by solving the URANS equations over the complete aircraft using the *elsA* structured CFD solver developed at ONERA. It has been used for many years for helicopter applications (Ref. 4) and more recently for specific Fenestron computations (Refs. 5, 6).

The 3D compressible URANS equations are solved using the two equations k-ω turbulence model (Refs. 13, 14) with the limiter proposed by Zheng and Liu (Refs. 15, 16) and the corrections of Kok (Ref. 17) and Menter (Ref. 18). The classical 2nd order centered scheme with scalar artificial viscosity (Ref. 19) and correction proposed by Martinelli and Jameson (Ref. 20) are used for the spatial discretization. The numerical scheme is implicit in time. The time integration is discretized by a second-order upwind three-time-level formula which is solved by using a Newton algorithm. The time step used for computations corresponds to an azimuthal blade displacement of 1 degree.

The geometry of the computed aircraft, displayed in Fig. 17, has been simplified with respect to the real helicopter: the lateral stabilizers have been removed and simplification have been done to the engines fairing geometry. The main rotor is modeled using a non-uniform actuator disk method, accounting for the load of the rotor over one revolution, included in two cylindrical blocks with approximately 16,000 cells. This steady state approximation of the main rotor reduces drastically the cost of the unsteady computation and allows densifying the computational grid in the tail rotor area. The Navier-Stokes background grid contains 85 blocks and 1.3 million cells. The grid in the main part of the fuselage is a relatively coarse grid and it has been decided to limit the number of cells outside the Fenestron duct for CPU time consumption reasons.

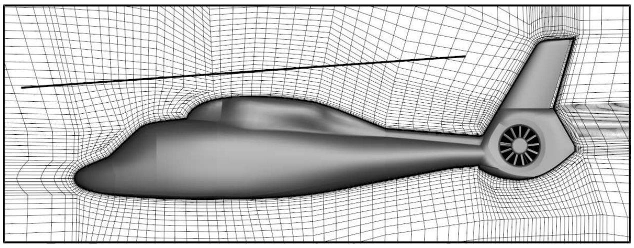

Fig. 17. Simplified geometry used for aerodynamic computations, actuator disk location and partial view of the computational grid

The geometry of the computed Fenestron, displayed in Fig. 18, is fully discretized with a fixed part consisting in the duct, the 3 arms (the one containing the drive shaft larger than the others), the hub and a rotating part consisting in the 13 equally spaced blades with blade tip gaps. Each of the 13 Fenestron blades is meshed with 3 blocks, containing approximately 110,000 cells. The blade motion during the CFD computations is taken into account thanks to the Chimera method developed at ONERA (Ref. 21) for helicopter configurations and consisting in the use of overlapping grids attached to the bodies in relative motion with respect to the Fenestron rotor blades as shown in Fig. 19. Interpolations are used to transfer the conservative and turbulent variables in the overlapping zone between the moving and fixed grids. The unsteady moving grid system contains 1.4 million cells. The complete grid system used for the CFD computations comprises 2.7 million cells.

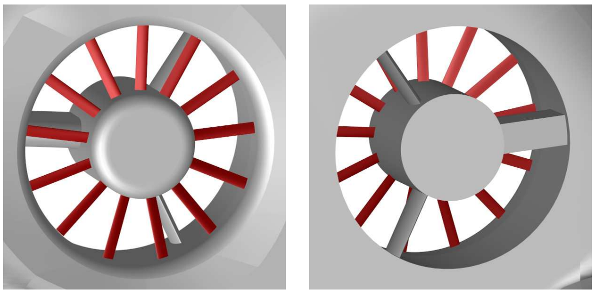

Fig. 18. Port and starboard views of the computed Fenestron Tail Rotor

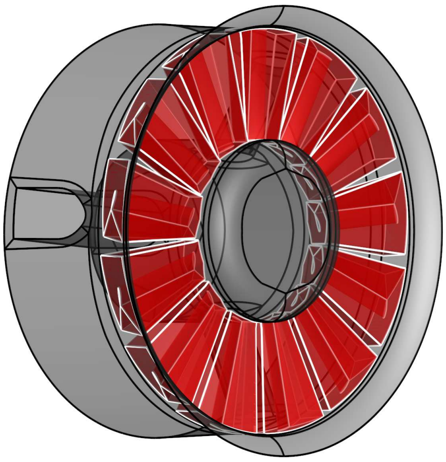

Fig. 19. Chimera grids around rotating blades

Aeroacoustic computations

The aeroacoustic computations are done by solving the Ffowcs Williams–Hawkings (Ref. 22) (called here after FW-H) equation formulated for solid or porous surfaces. The radiated noise is computed by solving the FW-H equation in the time domain using the KIM solver (Ref. 23) developed at ONERA.

The acoustic computations are made under the hypothesis that all acoustic sources are enclosed by the acoustic data surface so that the quadrupole term of the FW-H equation outside the closed surface can be neglected. No influence of solid surfaces, nor of the inhomogeneous inflow are accounted for outside the integral surface in the noise radiation computations. Furthermore, since the aerodynamic solutions provided by the CFD computations should be periodic in time, only discrete frequency noise is computed.

The sensitivity of the acoustic computations to the choice of the acoustic source surface used for the FW-H computations has been studied using several surfaces located at different sections in the shroud of the Fenestron. The first surfaces considered are the blade skins themselves (see Fig. 18) while the last surfaces considered are placed in such a way that they contain all acoustic sources inside the Fenestron and account for any reflection or diffraction occurring in the shroud (see Fig. 20).

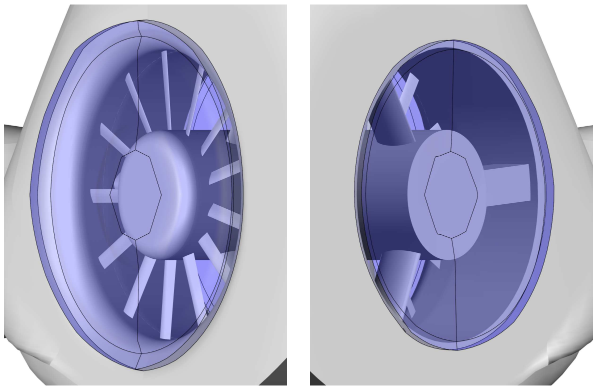

Fig. 20. Porous surfaces used for aeroacoustic computations based on the FW-H equations

Results and discussions

The flight condition chosen to compare numerical simulations against flight test data is a level flight at speed 150 kt. The simulation of a Fenestron fan-in-fin tail rotor at high-speed forward flight condition is very difficult for two main reasons: the first reason is that at high speed the tail fin provides a significant amount of the thrust required to counterbalance the main rotor torque and any slight variation of the helicopter sideslip angle will substantially increase or decrease the thrust to be produced by the Fenestron; the second reason is that since the Fenestron has to produce a slight amount of total thrust required to balance the main rotor, the mass flow through the Fenestron becomes very low so that flow separations occur at different locations inside the Fenestron duct (Refs. 4–6, 24).

The computation of this flight condition was done prior to the flight test so that the thrust to be provided by the Fenestron tail rotor in order to counterbalance the main rotor torque was estimated thanks to the HOST comprehensive code (Ref. 25). Several CFD computations with different blade pitch angles were necessary to match with the target thrust of the Fenestron. The computed wall pressure coefficient distribution over the full aircraft and main rotor actuator disk pressure distribution are displayed in Fig. 21. A close-up view of the wall pressure distribution inside the Fenestron is given in Fig. 22, some streamlines have also been plotted in this figure and highlight the position of the flow separation occurring in the upwind area of the Fenestron intake. This flow separation is attached to the wall over nearly half of the intake and is located in the blade tip crossing area. The computed pressures at 0.7 radius on the upper and lower sides of the blades are compared in Fig. 23 with the time synchronous averaging of the measured pressures on two different blades. The computed and experimental pressures exhibit similar behaviors over a tail rotor revolution but the CFD solution overestimates the pressure fluctuation. This discrepancy is most likely due to an overestimation of the target thrust to be provided by the Fenestron combined with differences between the experimental sideslip angle and the value used for the computation.

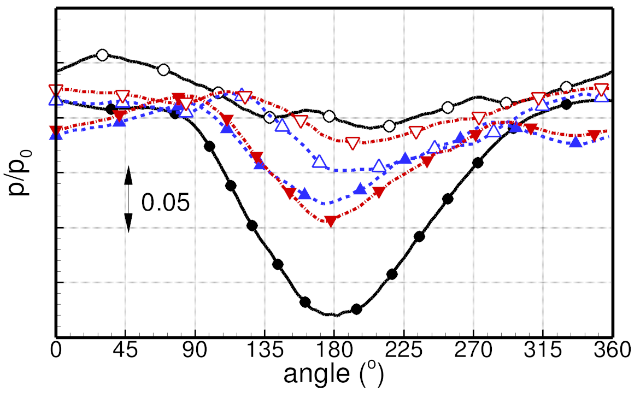

Fig. 23. Time evolution of unsteady blade pressure on suction (●, ▼, ▲) and pressure (○, ▽, △) sides at 0.7 R: present computation (——), pressure on blade n°2 (- - -) and n°4 (-·-) for a test flight at speed $V_{TAS} = 150$ kts

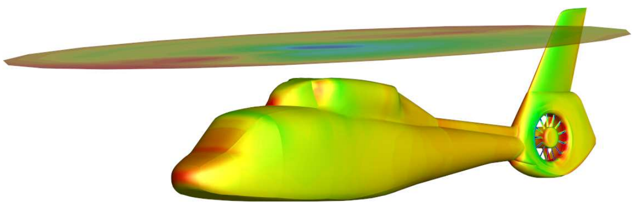

Fig. 21. Wall pressure coefficient distribution and actuator disk pressure distribution for a level flight computation at speed $V_{TAS} = 150$ kts

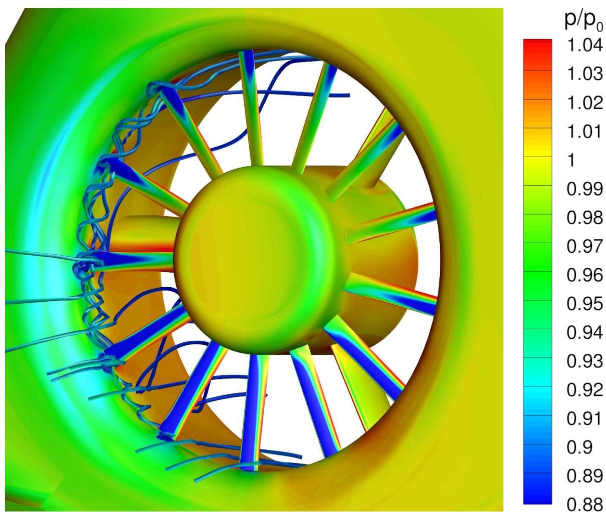

Fig. 22. Computed streamlines and wall pressure field in the Fenestron at level flight condition

As explained in the previous section, two aeroacoustic computations based on the Ffowcs Williams–Hawkings (FW-H) equation have been performed. The first one using unsteady blade pressures as input data for the solid surface formulation of the FW-H equation and the second one using the unsteady aerodynamic field on the porous surface encompassing the Fenestron inlet and outlet as input data for the porous surface formulation of the FW-H equation. The predicted overall sound pressure levels in a horizontal plane 150 m below the helicopter for both aeroacoustic computations are displayed in Fig. 24 and 25. The noise footprints in the CFD reference frame of the corresponding 4 flight tests at speed $V_{TAS} = 150$ kts are displayed in Fig. 26.

For the acoustic computation based on blade pressure data, the noisiest area of the footprint is located 250 m ahead of the aircraft and extends on a quarter of the map. Two other areas located 175 m downstream and ± 200 m at port side and starboard exhibit SPL lobes. Note that for this first computation, the maximum computed SPL in the noisiest area of the footprint is 10 dBA higher than the maximum SPL of the experiments.

For the acoustic computation based on the porous surface encompassing the Fenestron inlet and outlet, a significant part of the acoustic energy in the rotor plane is redistributed to the port and starboard directions meaning that this computation allows taking into account the masking and diffracting effects of the shroud. The ground footprint of this computation exhibit an SPL distribution in better agreement with respect to the experiments but there is still an area 250 m ahead the aircraft where the computed SPL are much higher than the measured SPL. Since it is located in an area upstream to advancing blade side of the tail rotor, the overestimation of the SPL ahead the aircraft is most likely linked to the overestimation of the blade pressure fluctuation on the blades.

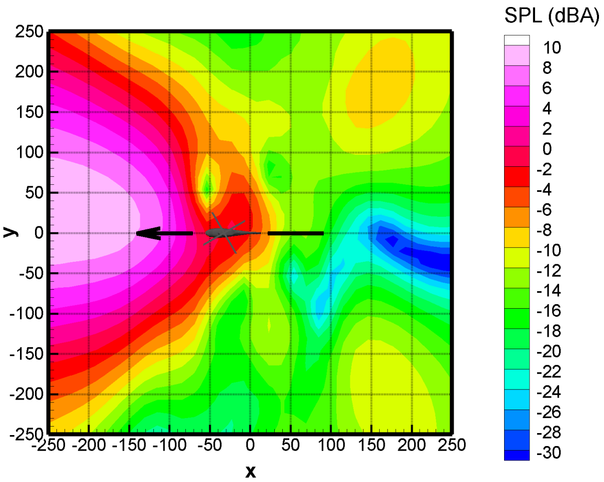

Fig. 24. Fenestron noise footprint computed using unsteady blade pressure as input data for the solid formulation of the FW-H equation

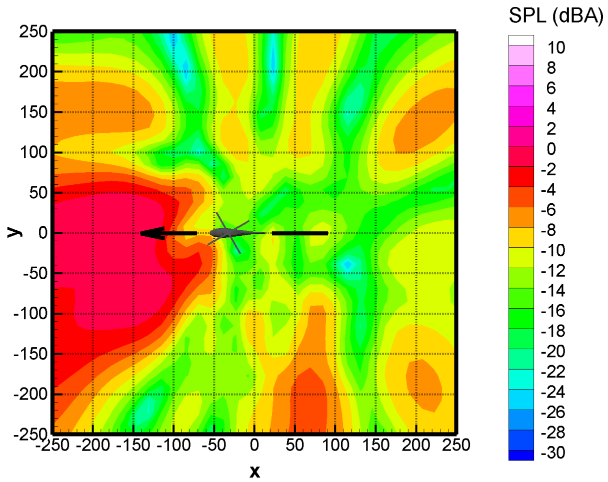

Fig. 25. Fenestron noise footprint computed using unsteady aerodynamic field on the fluid surface defined in Fig. 20 as input data for the porous formulation of the FW-H equation

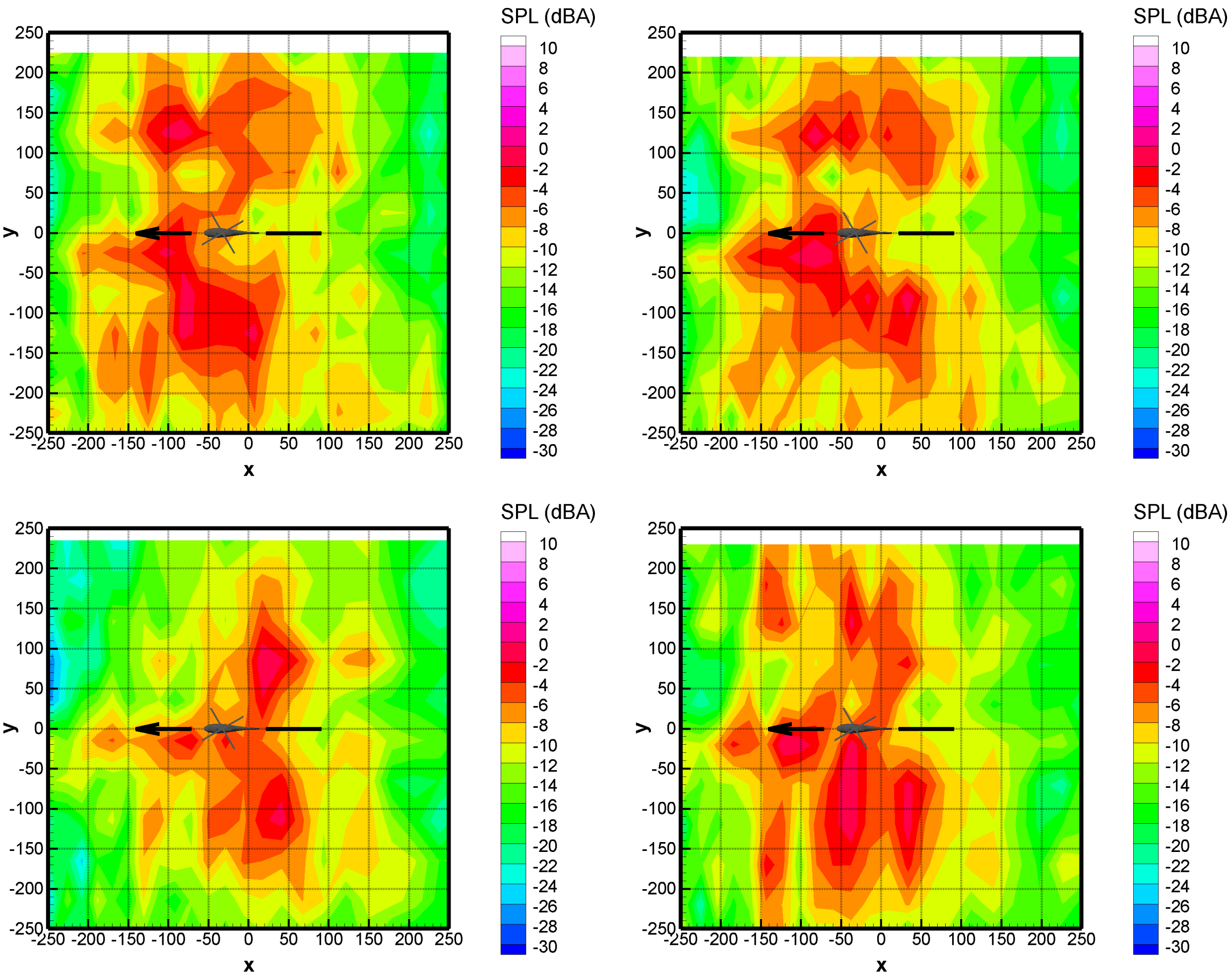

Fig. 26. Fenestron noise footprint computed using microphone measurements of 4 different flight test points corresponding to a level flight at $V_{TAS} \simeq 150$ kts

Actually, the experimental blade pitch angle has been measured during the second flight test campaign to settle this uncertainty and the effective value is clearly lower than the one used in the computations done prior to the flight tests. New aerodynamic and aeroacoustic computations will be performed using the experimental value and compared against the experiments. It is expected that the flow physics inside the duct will be rather different than the one presented here and it should be as well different regarding the radiated noise. Finally, note that for the second acoustic computation of the present study, the maximum computed SPL is only 1 dBA higher than the maximum SPL of the experiments.

Conclusion and future work

One of the objective of the Fenestron noise study was to provide an experimental database for the validation of aerodynamic and aeroacoustic computations of a Fenestron shrouded tail rotor. Two flight test campaigns involving a SA 365 N Dauphin 2 helicopter of the French Flight Test Center (CEV) have been carried out and have provided extensive aerodynamic and acoustic databases.

The experimental setup used for in-flight aerodynamic and acoustic measurements as well as the plan and methodology used for ground measurements have be presented. In order to obtain data directly comparable with CFD and CAA computations a specific post-processing has been applied to the experimental measurements.

The simulation tools developed at ONERA for Fenestron aerodynamic and aeroacoustic simulations have been presented and applied to the computation of flight condition corresponding to a level flight at speed 150 kts.

The comparison of the computation results against the experiment has shown that the complex physics of the flow in the Fenestron and its acoustic radiation is partly captured by the CFD and CAA methods but it also pointed out that the trim of the Fenestron rotor used in the computations was different to the effective trim in the flight test. This disparity on the Fenestron blade pitch angle is most likely responsible of the discrepancy observed on the computed and measured blade pressures and could also affect strongly the computed noise directivities.

New computations using the experimental trim are currently underway to answer this question. Once the computed blade pressures will fit the experimental values, a grid convergence study will be carried out by using a finer computational grid for the CFD computation in order to ensure that the acoustic sources propagate correctly to the porous surfaces used for the noise radiation computations.

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